Aggregate Demand and Aggregate Supply-Solutions

ECON 101: Macroeconomics

Exercise 1: Calculating AD and AS Equilibrium

The aggregate demand (AD) and short-run aggregate supply (SRAS) for an economy are given by the following equations:

$$AD: \quad Y = 5000 - 200P$$
$$SRAS: \quad Y = 2500 + 100P$$

where Y is real GDP and P is the price level.

Questions:

1. Find the equilibrium price level (P^*) and equilibrium output (Y^*).
2. Suppose consumer confidence decreases, shifting AD to $Y = 4800 - 200P$. Find the new equilibrium.
3. Illustrate the initial and new equilibrium graphically using AD and AS curves.

Solutions:

1. To find equilibrium, set AD equal to SRAS:

$$5000 - 200P = 2500 + 100P$$
$$2500 = 300P$$
$$P^* = \frac{2500}{300} \approx 8.33$$

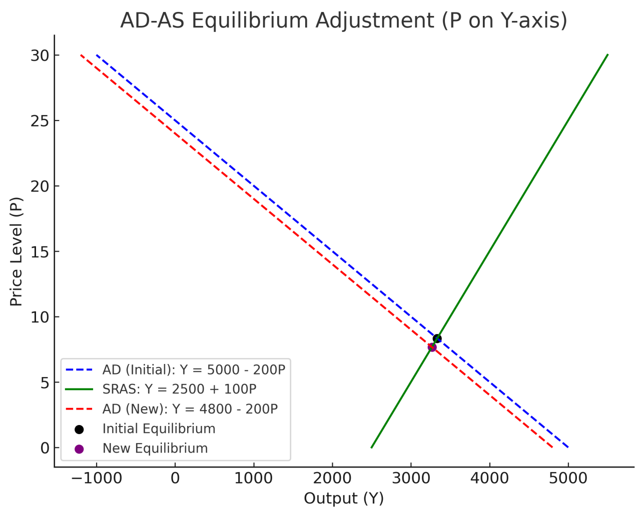
   Substituting P^* into either equation to find Y^*:

$$Y^* = 5000 - 200(8.33) \approx 3333.33$$

2. For the new AD equation $Y = 4800 - 200P$, solve:

$$4800 - 200P = 2500 + 100P$$
$$2300 = 300P$$
$$P^* = \frac{2300}{300} \approx 7.67$$

$$Y^* = 4800 - 200(7.67) \approx 3266.67$$



Exercise 2: Graphing the Effects of a Negative Supply Shock

Consider an economy in equilibrium at $Y = 5500$ and $P = 50$. A supply shock (e.g., an oil price increase) shifts SRAS leftward from:

$$SRAS_1: \quad Y = 3000 + 50P$$
$$SRAS_2: \quad Y = 2800 + 50P$$

Solutions:

Macroeconomics Problem: Equilibrium after Supply Shock

Given:

- Initial equilibrium: $Y = 5500,\ P = 50$
- Initial SRAS (SRAS1): $Y = 3000 + 50P$
- New SRAS (SRAS2): $Y = 2800 + 50P$
- Aggregate Demand (AD): $Y = 8000 - 50P$

Determine the new equilibrium price level and output.

We set the aggregate demand (AD) equal to the new short-run aggregate supply (SRAS2) curve:

$$AD = SRAS2$$

$$8000 - 50P = 2800 + 50P$$

Solving for P:

$$8000 - 2800 = 50P + 50P$$

$$5200 = 100P$$

$$P = \frac{5200}{100} = 52$$

Now, substitute $P = 52$ into the AD curve to find Y:

$$Y = 8000 - 50(52) = 8000 - 2600 = 5400$$

Thus, the new equilibrium price level is $P = 52$ and the new equilibrium output is $Y = 5400$.

llustrate the shift in SRAS using a graph.

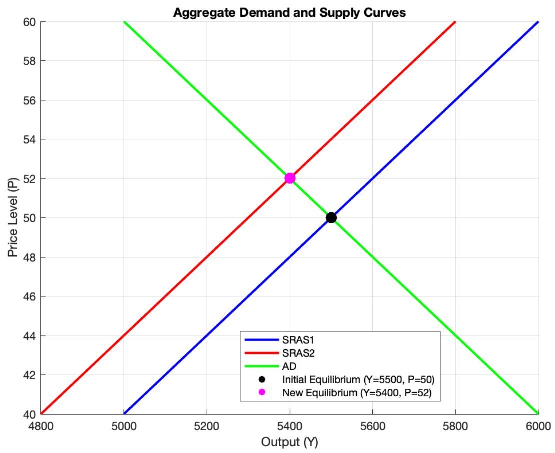
To illustrate the shift in SRAS:

- Initially, SRAS1 is $Y = 3000 + 50P$, and the economy is at $Y = 5500$ and $P = 50$.
- After the supply shock, the new SRAS curve is $Y = 2800 + 50P$.
- The AD curve is $Y = 8000 - 50P$.

The AD curve remains unchanged, and the SRAS curve shifts leftward from SRAS1 to SRAS2, reflecting a decrease in output at each price level. The new equilibrium occurs at $P = 52$ and $Y = 5400$.

What happens to inflation and unemployment after this shock?

- **Inflation:** The price level rises from $P = 50$ to $P = 52$, indicating an increase in inflation as the supply shock (e.g., oil price increase) pushes up costs, leading to higher prices.
- **Unemployment:** Output decreases from $Y = 5500$ to $Y = 5400$, implying a decline in economic activity, which leads to higher unemployment as firms reduce production and need fewer workers.



If the central bank responds by increasing the money supply, how would AD shift?

If the central bank increases the money supply, the AD curve would shift to the right. This shift occurs because:

- An increase in the money supply lowers interest rates, which stimulates investment and consumption.
- As a result, demand for goods and services increases, shifting the AD curve to the right.
- This would raise output and potentially increase the price level, helping to offset the negative impact of the supply shock on the economy.

Exercise 3: Fiscal Policy and AD Shifts

Suppose the government implements an expansionary fiscal policy by increasing spending by $200 billion. The marginal propensity to consume (MPC) is 0.8.

Solutions:

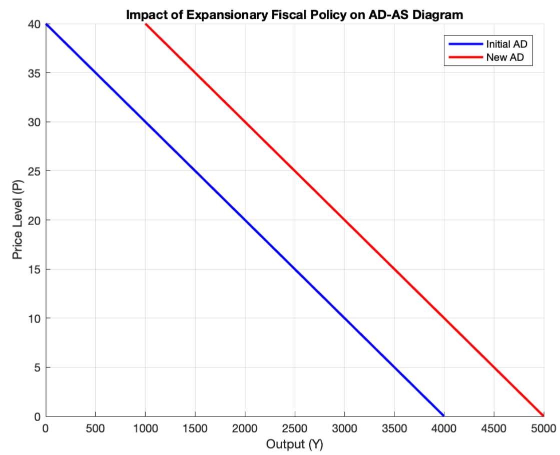
1. Compute spending multiplier:

$$\text{Multiplier} = \frac{1}{1 - 0.8} = 5$$

2. Compute total GDP increase:

$$\Delta Y = 5 \times 200 = 1000$$

3. New AD equation: $Y = 5000 - 100P$



Exercise 4: Long-Run Adjustment to a Recession

An economy is experiencing a **recessionary gap** with actual GDP at 2700 and full-employment GDP at 3000. Assume wages are **sticky in the short run** but adjust over time.

Solutions:

1. Graph the initial gap.
2. Over time, SRAS shifts right as wages adjust.
3. Government intervention can speed up recovery via fiscal/monetary policy.
4. Classical economists advocate for self-correction, Keynesians favor active policy.